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- The diagram compares the structure of a plant cell (*Chlorella*) and an animal cell (*Amoeba*).
- Chlorella (Plant Cell):**
- A:** Points to the thick, rigid **cellulose cell wall**.
 - B:** Points to the large, central **vacuole**.
- Amoeba (Animal Cell):**
- nucleus:** A large, dark, spherical structure.
 - plasma membrane:** The outer boundary of the cell.
 - contractile vacuole:** A small, clear, spherical structure.

- [1]

organelle **B**:

- (i) Use your knowledge of osmosis to explain why contractile vacuoles are necessary for the survival of the *Amoeba*. [4]



- (ii) Identify **two** *other* organelles which must be present in large numbers in the *Amoeba* in order for this process to occur. Explain your answer for each organelle. [4]

Organelle 1

Explanation

Organelle 2

Explanation

- (c) Although *Chlorella* lives in the same habitat as *Amoeba*, it does not possess a contractile vacuole. Describe and explain how the structure of the cell wall of *Chlorella* allows it to survive in this environment. [4]

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- (d) Bacteria, such as *Cyanophyceae* also inhabit fresh water ponds. Describe **two** differences between the structure of *Chlorella* and *Cyanophyceae*. [2]

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